Ride Lanka - AI Chatbot Knowledge Base 1. About Sri Lanka Sri Lanka is an island nation in the Indian Ocean known for its rich cultural heritage, scenic landscapes, and tourism attractions. Popular destinations include Ella, Galle, Mirissa, Hikkaduwa, and Kandy. The country offers beaches, mountains, wildlife, and historical sites. 2. Tourism in Sri Lanka Sri Lanka attracts millions of tourists annually due to its biodiversity, UNESCO heritage sites, and cultural festivals. Tourism contributes significantly to the economy and supports local businesses. 3. About Ride Lanka Platform Ride Lanka is a web-based bike rental platform that connects tourists with local bike vendors. Users can browse bikes, book rentals, and explore tourist destinations. Vendors can list bikes, and admins approve listings. 4. Bike Rental Information Users can rent scooters and gear bikes. Booking requires selecting dates, location, and bike type. Payments can be made via online methods or bank transfer. 5. Chatbot Use Cases The chatbot can answer questions such as: - How to rent a bike? - What are the best places to visit in Sri Lanka? - Which bike is suitable for Ella? - How to make payments? - What documents are required? 6. FAQs for Chatbot Q: How can I book a bike? A: Browse bikes, select dates, and confirm booking. Q: Can I cancel a booking? A: Yes, based on vendor policy. Q: What payment methods are available? A: Online payment or bank transfer. 2. Bike Rental Questions Q6: How can I rent a bike? A: Browse bikes, select dates, choose location, and confirm booking. Q7: What types of bikes are available? A: Scooters and gear bikes. Q8: Which bike is best for beginners? A: Scooters are recommended for beginners. Q9: Which bike is best for mountains? A: Gear bikes are best for hill areas like Ella. Q10: Can I rent multiple bikes? A: Yes, based on availability. 3. Payment Questions Q11: What payment methods are available? A: Online payment or bank transfer. Q12: How do I confirm bank transfer? A: Upload your payment slip for admin verification. Q13: Is payment secure? A: Yes, the system uses secure verification. 4. Booking Questions Q14: Can I cancel a booking? A: Yes, depending on vendor policy. Q15: Can I modify my booking? A: Yes, before confirmation. Q16: How is rental cost calculated? A: Days × price × quantity. 5. Vendor Questions Q17: How do I become a vendor? A: Register as vendor and submit details. Q18: How are bikes approved? A: Admin reviews and approves listings. Q19: Can vendors update bike details? A: Yes, anytime. 6. Tourism Questions Q20: What are the best places in Sri Lanka? A: Ella, Galle, Mirissa, Hikkaduwa, Kandy. Q21: What is Ella famous for? A: Mountains, waterfalls, Nine Arches Bridge. Q22: What is Galle famous for? A: Dutch Fort and beaches. Q23: What is Mirissa famous for? A: Whale watching and beaches. Q24: What is Hikkaduwa famous for? A: Coral reefs and surfing. 7. Travel + Bike Recommendations Q25: Best bike for Ella? A: Gear bike. Q26: Best bike for Galle? A: Scooter. Q27: Best bike for Mirissa? A: Scooter. Q28: Best bike for long travel? A: Gear bike. Q29: Best bike for city travel? A: Scooter. 8. Safety Questions Q30: Do I need a license? A: Yes, a valid driving license is required. Q31: Is helmet provided? A: Yes, vendors provide helmets. Q32: Is insurance included? A: Depends on vendor policy. 9. System Features Q33: Can I see reviews? A: Yes, only approved reviews are shown. Q34: Can I leave a review? A: Yes, after completing a booking. Q35: How are recommendations generated? A: Based on location, ratings, and popularity. 10. Advanced Questions Q36: How does AI recommend bikes? A: It analyzes user behavior, ratings, and location data.

Q37: What happens if no bikes are available? A: Try changing location or dates. Q38: Can I filter bikes? A: Yes, by price, type, and location. Q39: Can vendors reject bookings? A: Yes, based on availability. Q40: Can I book for future dates? A: Yes. 11. Extra Smart Questions Q41: Is Ella good for biking? A: Yes, especially for adventure rides. Q42: Is Sri Lanka safe for tourists? A: Yes, it is a popular tourist destination. Q43: What is the best time to visit Sri Lanka? A: December to April for west/south coast. Q44: Can I travel between cities by bike? A: Yes, but use gear bikes for long distances. Q45: Are roads good for bikes? A: Most tourist areas have good roads. 12. Edge Cases Q46: What if my payment fails? A: Retry or use another method. Q47: What if vendor cancels? A: You can choose another bike. Q48: What if bike is damaged? A: Contact vendor immediately. Q49: Can I extend rental time? A: Yes, based on availability. Q50: Can tourists use this system easily? A: Yes, it is designed for user-friendly experience. Part 1: Paragraphs (For Chatbot Knowledge Base / General Context) 1. Exploring the Southern Coast by Motorbike The Southern Province of Sri Lanka offers one of the most scenic coastal rides in the world. Renting a scooter or motorbike is the best way to explore popular destinations like Galle Fort, Unawatuna, Mirissa, Weligama, Hiriketiya, and Tangalle. The main coastal route is the A2 highway (Galle Road), which provides easy access to beautiful beaches, surfing spots, and local cafes. Traveling by bike gives tourists the freedom to discover hidden gems like Secret Beach in Mirissa or the rope swings in Dalawella at their own pace. 2. Legal Requirements and Driving Licenses To legally ride a motorbike in Sri Lanka, foreign visitors must have a valid driving license. An International Driving Permit (IDP) alone is not sufficient; it must be endorsed by the Automobile Association of Ceylon (AAC) in Colombo. Alternatively, visitors can obtain a temporary Sri Lankan driving license from the Department of Motor Traffic (Werahera) by presenting their home country's valid driving license. Police checkpoints are common in the Southern Province, so riders must always carry their passport, valid license, vehicle registration, and insurance documents. 3. Road Safety and Traffic Rules Safety is a top priority when riding in Sri Lanka. It is a strict legal requirement for both the rider and the passenger to wear standard helmets at all times. Motorbikes and scooters are strictly prohibited on the Southern Expressway (E01); all bike travel must be done on normal roads like the A2 coastal road. Riders should be cautious of local traffic, including fast-moving private buses, tuk-tuks, and stray animals like dogs or peacocks crossing the roads. It is highly recommended to ride during daylight hours and avoid night riding in poorly lit areas. Part 2: Q&A Pairs (For Intent and Response Training) Q: Do I need an international license to rent a bike? A: You need an International Driving Permit (IDP) with a motorcycle endorsement, but it must be verified and stamped by the Automobile Association of Ceylon (AAC). Alternatively, you can get a temporary Sri Lankan license using your home country's driving license. Q: Can I ride the scooter on the highway to get to the South faster? A: No. Motorbikes and scooters are strictly prohibited on the Southern Expressway (E01). You must use the scenic A2 coastal road (Galle Road) to travel between towns. Q: Do I have to wear a helmet? A: Yes, it is a strict law in Sri Lanka. Both the rider and the passenger must wear a helmet at all times. Failure to do so will result in a police fine. Q: Where can I get fuel for the bike? A: There are plenty of petrol stations (Ceypetco, IOC, and Sinopec) along

the main A2 coastal road in towns like Galle, Weligama, and Matara. We recommend carrying some cash, as smaller stations might not accept credit cards. Q: What are the best places to visit in the South on a bike? A: The South is perfect for biking! Popular spots include Galle Fort for sunsets, Weligama and Hiriketiya for surfing, Mirissa for whale watching, and Unawatuna for relaxing beach vibes. Q: Can I attach my surfboard to the rental bike? A: Yes! Many of our bikes come with an attached surfboard rack upon request. Please let us know when booking if you need one, especially if you are heading to surfing hotspots like Weligama or Midigama. Q: What should I do if the bike breaks down or I get a flat tire? A: Don't worry! There are small local repair shops (called "wadu maduwa" or tire shops) in almost every village. If it's a major breakdown, contact our customer support number immediately, and we will arrange roadside assistance or a replacement bike. Q: What happens if I get stopped by the police? A: Police checkpoints are normal. Stay calm, be polite, and show them your valid driving license (endorsed IDP or local temporary license), passport, and the bike's registration and insurance documents that we provide. 1. AI-Powered Bike Recommendations Our platform features an advanced AI-assisted bike recommendation engine to help you find the perfect ride. By analyzing your travel location, trip duration, budget, terrain, and past rental history, the AI suggests the most appropriate bike type for your journey. For example, the system will recommend a lightweight scooter for coastal rides, while suggesting a higher-torque motorbike for hilly terrains like Ella or Nuwara Eliya. 2. Fair and Transparent Dynamic Pricing To prevent overcharging and ensure transparent rates, our platform utilizes a machine learning-based dynamic pricing prediction model. This system estimates reasonable daily rental prices by evaluating variables such as the current season, market demand, location, and the specific type of bike. This guarantees that tourists always pay a fair price, especially during peak seasons. 3. Real-Time Route Safety and Navigation Tourist safety is a core priority of our platform. We have integrated Google Maps and Weather APIs to provide real-time safety warnings and route optimization. While you travel, the system actively alerts you about wet or slippery roads, accident-prone areas, high traffic, and steep hilly terrains, ensuring a secure and confident riding experience. 4. Verified Vendors and Secure Bookings All bike listings on the platform go through a strict verification process. Vendors must create an account and submit their bikes for an approval workflow. Administrators carefully review and approve these listings before they become visible to the public, ensuring you always rent from reliable and trusted local businesses. Part 2: Q&A Pairs (For Intent and Feature Training) Q: How do I know which bike is best for my trip? A: Our AI recommendation engine will suggest the perfect bike for you!. Just enter your destination, trip duration, and budget, and the system will recommend whether a scooter or a geared motorbike is best suited for your route's terrain. Q: Are the rental prices fixed, or do they change? A: We use an AI-driven dynamic pricing system to ensure you always get a fair and transparent rate. The prices are calculated based on current market demand, the season, and the type of bike, protecting you from being overcharged. Q: Is it safe to ride a bike in unfamiliar areas of Sri Lanka? A: Yes, our platform includes a smart route and safety alert system. You will receive live warnings about bad weather, slippery roads, steep terrains, and accident

hotspots along your route to ensure a safe journey. Q: How can I find available bikes near my hotel? A: You can easily use our GPS-based search feature powered by Google Maps to locate nearby available bikes in real-time. Q: Are the vendors and bikes on the platform trustworthy? A: Absolutely. All vendors and their bikes go through an administrative approval workflow. Our admins give bike listings the green light before they become visible to the public to ensure quality and safety. Q: How can I pay for my bike rental? A: We support secure online payments via integrated payment gateways like Stripe and PayPal. Alternatively, we also support bank transfer payments where you can upload your payment slip for verification. Q: What happens if I want to leave a review after my trip? A: We encourage it! Our platform has a verified review and rating management system. Your feedback helps other tourists make informed decisions and helps vendors improve their services. ## 8) Notification System Real-time: - Socket event server->client: new_notification, pending_notifications - Socket event client->server: notification_read Delivery behavior: - Save notification in DB - If recipient socket connected, emit in real-time - Optionally send email via Nodemailer - Unread notifications can be fetched later (offline support) Email templates: - HTML layout includes title/message and optional CTA button. - Booking received includes link to vendor bookings page. ## 7.5 Review Rules - Order must be completed. - Must belong to authenticated user. - Duplicate review for same user+order+product is blocked. Part 3: Paragraphs (Vendor Dashboard, Destinations & Pricing) 1. Vendor Dashboard and Business Analytics The platform provides a dedicated, easy-touse dashboard specifically for bike owners and rental vendors. Through this dashboard, vendors can register their business, add new bike listings with photos, and set flexible rental rates such as daily, weekly, or monthly charges. To help vendors grow their business, the system includes an AI-based analytics tool that shows booking frequencies, peak rental times, and popular tourist routes. Furthermore, a predictive maintenance feature tracks bike usage and automatically alerts vendors when a bike needs servicing, ensuring all vehicles remain safe and in top condition. 2. Must-Visit Destinations in the Southern Province The Southern coast of Sri Lanka is one of the most popular regions for motorbike exploration. Tourists highly prefer Hikkaduwa, Mirissa, and Unawatuna for their beautiful beaches, coral reefs, and whalewatching experiences. Weligama and Midigama are world-renowned surfing hotspots, perfect for riders looking to catch some waves. For history and architecture lovers, a ride to the historic Galle Fort is highly recommended. If tourists plan to travel slightly inland towards Ella, the terrain changes to steep hills and tea plantations, requiring a more powerful geared motorbike rather than a standard coastal scooter. 3. Estimated Rental Pricing Guide (Sri Lanka) Rental prices in Sri Lanka vary based on the type of bike, the duration of the rental, and the tourist season. For a standard scooter (such as a Honda Dio or TVS Ntorq), the daily rental rate typically ranges from LKR 2,000 to LKR 3,500. For larger, geared motorbikes (such as a Honda XR 150 or Royal Enfield) that are better suited for hilly terrains or long-distance touring, the daily rate ranges from LKR 4,500 to LKR 7,000. Our platform uses an AI-driven dynamic pricing model to ensure these rates are always fair, preventing tourists from being overcharged during peak seasons while optimizing revenue for vendors. Part 4: Q&A Pairs (For Intent and Feature

Training) Q: How much does it cost to rent a scooter for one day in the South?A: A standard scooter typically costs between LKR 2,000 and LKR 3,500 per day. However, our system uses dynamic pricing, so the exact rate might adjust slightly based on current demand and the season to guarantee you get a fair deal. Q: Do I get a discount if I rent the bike for a whole week?A: Yes! Vendors on our platform can set specific rates for longer durations. If you book a bike for a week or a month, the total cost calculated will usually include a long-term discount compared to the standard daily rate. Q: How do I, as a vendor, know when my bikes need to be serviced?A: Your vendor dashboard comes with a smart predictive maintenance feature. It tracks the usage and mileage of your rented bikes and will send you an alert when it's time for a service, helping you reduce downtime and keep your customers safe. Q: I want to go surfing. Where are the best places to ride my rented bike?A: The Southern Province is perfect for surfing! You should definitely ride down to Weligama, Midigama, or Hiriketiya. Many of our vendors in these areas can even provide a scooter with a surfboard rack attached. Q: Can I ride a regular scooter to Ella?A: While it is possible, Ella has very steep and hilly terrain. For your safety and comfort, our AI recommendation engine strongly suggests renting a geared motorbike with higher torque for trips into the hill country. Q: I am a vendor. How do I track my earnings and bookings?A: You can log into your vendor dashboard at any time to view your complete order history, track ongoing rentals, and use our analytics tools to see your peak booking times and customer preferences.

1. Additional Traffic & Safety Rules

The knowledge base correctly states that riders need an AAC-endorsed IDP or a temporary local license. It is a strict law for both the rider and passenger to wear standard helmets at all times. Motorbikes and scooters are strictly prohibited on the Southern Expressway (E01). All bike travel must be done on normal roads like the A2 coastal road. However, the chatbot also needs to know specific speed limits and emergency protocols.

- **Speed Limits:** In city and built-up urban areas, the speed limit for motorcycles is **40 km/h**. On open roads outside the city (non-built-up areas), the limit is **60 km/h**.
- **Emergency Contact Numbers:**
    - Police Emergency: **118** or **119**
    - Tourist Police Hotline: **011 242 1052**
    - Suwa Seriya Ambulance Service: **1990**
    - Accident Service General Hospital (Colombo): **011 269 1111**

- **Fuel Station Details:** There are plenty of petrol stations like Ceypetco, IOC, and Sinopec along the main A2 coastal road. Smaller stations might not accept credit cards, so carrying cash is recommended. Tourists should also know to specify "Petrol 92" (standard) or "Petrol 95" (premium) when filling up, as attendants pump the fuel for you in Sri Lanka.

2. Hidden Gems & Southern Province Tourist Spots

The South is perfect for biking. Popular destinations include Ella, Galle, Mirissa, Hikkaduwa, and Kandy. Expanding on the Southern Province spots will make the chatbot an excellent virtual tour guide.

- **Off-the-Beaten-Path Beaches:** Traveling by bike gives tourists the freedom to discover hidden gems. Instead of just main beaches, add **Secret Beach** in Mirissa and **Jungle Beach** in Unawatuna. These are highly sought after by tourists looking for quieter spots.
- **Wildlife & Nature:** Mirissa is famous for whale watching. Beyond this, suggest **Yala National Park**, **Bundala National Park**, and **Lunugamvehera National Park** near Tissamaharama for leopard and elephant safaris. For bird watching and boat safaris, the **Madu Ganga River** in Balapitiya offers fantastic mangrove tours.
- **Cultural & Authentic Experiences:**
    - **Stilt Fishing:** A traditional practice tourists love to photograph, found mostly around the waters of **Koggala** and **Ahangama**.
    - **Ambalangoda Mask Museum:** A great coastal stop to see traditional Sri Lankan mask carving and learn about local folklore.
    - **Galle Fort:** A highly recommended historic and architectural site. It is famous for sunsets.
    - **Rumassala Hill:** Located near Galle, offering stunning ocean views, a peace pagoda, and local myths tying it to ancient legends.

3. Expanded Bike Details & Recommendations

The system uses an AI recommendation engine to suggest bikes based on destination, trip duration, budget, and terrain. Scooters are recommended for beginners and coastal rides. Gear bikes are best for hill areas like Ella.

- **Bike Capacity (CC) Rules:** The chatbot should inform users that motorcycles with an engine capacity over 450cc are generally restricted by law in Sri Lanka, meaning the vast majority of rental bikes will be under 250cc.
- **Surfing Logistics:** The Southern Province is perfect for surfing, especially spots like Weligama, Midigama, and Hiriketiya. Bikes can come with an attached surfboard

rack upon request. The chatbot should advise users to ensure the bike they book is a standard scooter (like a Honda Scoopy or Dio), as it is much easier to attach racks to these than to heavy gear bikes.

- **Maintenance & Damage Protocol:** If a bike is damaged, users must contact the vendor immediately. For minor issues like a flat tire, there are small local repair shops (called "wadu maduwa" or tire shops) in almost every village. For major breakdowns, customer support will arrange roadside assistance or a replacement bike. It's useful to train the bot to advise tourists *not* to attempt engine repairs themselves to avoid voiding vendor agreements.

- **Dynamic Pricing:** The platform utilizes an AI-driven dynamic pricing model to ensure fair rates based on season, demand, location, and bike type. For a standard scooter, the daily rental rate typically ranges from LKR 2,000 to LKR 3,500. For larger, geared motorbikes, the daily rate ranges from LKR 4,500 to LKR 7,000.

4. General Sri Lankan Context (Weather & Routing)

- **Seasonal Routing:** The best time to visit the west/south coast is December to April. The chatbot should also inform tourists that during the off-season (May to October), the East Coast (like Arugam Bay for surfing) is the better destination, but they will definitely need a long-distance gear bike to get there from the South.

- **Smart Safety System Integration:** When users ask about road safety, the chatbot can emphasize your platform's smart routing that uses Google Maps and Weather APIs. The system actively alerts riders about wet or slippery roads, accident-prone areas, high traffic, and steep hilly terrains.

1. Key Tourist Places & Zones

To help your AI recommend the right bikes for the right locations, it needs to understand the distinct tourism zones in Sri Lanka:

- **The Southern Coastal Belt:** The South coast is one of the most popular travel regions in the country. It is highly sought after for its golden beaches, surfing, and vibrant beach entertainment events. Because tourists frequently start or end their trips at these beach destinations, this area experiences high, concentrated demand for local transport.

- **The Cultural Triangle:** This region attracts tourists interested in Sri Lanka's rich history and cultural heritage. Key sites include the ancient city of Anuradhapura, the famous Sigiriya rock fortress, and the ancient city of Polonnaruwa.

- **The Central Highlands (Hill Country):** Destinations like Kandy offer beautiful attractions, such as scenic walks around Kandy Lake and the historically significant Temple of the Tooth. Other notable locations include Adam's Peak (Sri Pada) for spiritual journeys and the Ambuluwawa Tower, a unique hilltop sanctuary.

- **The East & North:** Trincomalee on the east coast is famous for its pristine beaches and natural harbor. In the north, Jaffna offers a completely different experience, known for its rich Tamil culture and unique local food.

2. Seasonality and Peak Demand Areas

Understanding seasonality allows a smart platform to predict when and where bikes will be needed most:

- **Southern Peak:** The Southern and Western coasts experience peak demand during their dry season, driven heavily by international tourists seeking beach holidays and nightlife events.
- **Eastern Shift:** Expanding tourism to the Eastern Province helps extend Sri Lanka's overall tourism seasonality by an additional two months because the east has a different rainy season than the south. A smart bike rental platform could use this data to advise vendors to shift their fleets eastward during the southern off-season to maintain steady revenue.

3. How Tourism Affects Local Areas

Tourism drastically changes the local landscape. Understanding these impacts can help your platform promote responsible travel:

Positive Economic Impacts:

- **Economic Driver:** Tourism is a major economic engine for Sri Lanka, historically accounting for around 5% of the country's GDP and generating billions in foreign exchange earnings.
- **Small Business Growth:** The influx of tourists directly supports the development of local small businesses, including guesthouses, street vendors, and independent transport providers.
- **Job Creation:** The industry provides widespread employment opportunities for local communities, ranging from hotel staff and guides to drivers and event organizers.

Infrastructure Challenges:

- While tourism encourages the development of roads and public facilities, inadequate infrastructure remains a major issue. Roads to popular destinations are sometimes in poor condition, and public transport is not always reliable or comfortable for travelers. This specific challenge highlights exactly why a platform offering reliable, independent bike rentals is so valuable to tourists.

Negative Environmental and Social Strains:

- **Environmental Degradation:** High volumes of tourists can harm the environment, particularly in heavily visited areas like the southern coastal belt, where overdevelopment has disrupted local ecosystems. Increased tourism activity can lead to soil erosion, deforestation, groundwater depletion, and increased waste generation.
- **Pollution:** Popular tourist zones often suffer from increased air pollution from transportation, water pollution from wastewater, and noise pollution from activities and events.
- **Community Disruption:** While beach entertainment events (like club parties and kite festivals) draw enormous crowds and boost the economy, they can also cause negative socio-cultural impacts. These include the disruption of traditional ways of life, social conflicts between locals and tourists, and issues related to crime, drugs, and the commercialization of cultural heritage.